Artificial intelligence advancements in procurement: transforming organizational buying behavior

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Abstract

Purpose – The purpose of the paper is to discuss the potential of AI to impact organizational buying behavior. Potential impacts on organizational communication, Organizational Buying Center (OBC) structure and dynamics are discussed.

Design/methodology/approach – This research is conceptual in nature and uses narrative literature review to develop testable propositions derived from a technology overview and incorporates existing organizational buying behavior theory.

Findings – The article's conclusion suggests that significant changes are likely due to the adoption of AI. The nature of organizational buying is anticipated to undergo a shift toward increased reliance on AI-generated input. Additionally, it is expected that the size of OBCs may decrease, with reduced vertical and lateral involvement, while promoting greater coordination and less conflict among members.

Originality/value – This paper aims to conceptualize the effects of AI technology adoption on organizational buying situations and the structure and dynamics of OBCs. For practitioners, this conceptualization may equip them to navigate the impacts of the AI technological advancements effectively.

Keywords Efficiency, Effectiveness, Artificial intelligence, Organizational buying behavior

Paper type Conceptual paper

Introduction

The growth of Artificial Intelligence (AI) **already has had an impact on our everyday lives** well as on the **selling** behavior of many companies (McClure *et al.*, 2024). Researchers in marketing focus less on the technologies of AI and more on the potential of AI to impact business practices, such as interacting with customers, and suppliers, automating business processes and gain insights from data for personalization and product/service development (Davenport *et al.*, 2020). According to a survey conducted by Forbes 64% of businesses believe that AI will increase their productivity and 60% of businesses believe that AI will improve their customer relationships (Forbes, 2023). The speed of AI technological development was driven by the advances in computer hardware and software and the explosion of data sources from the Internet and social media (Brynjolfsson and McAfee, 2017). Academic researchers investigated how advertisers use AI to influence the consumer decision-making process (Kietzmann *et al.*, 2018), how AI is reshaping retailing (Shankar, 2018), sales (McClure *et al.*, 2024; Syam and Sharma, 2018), other aspects of marketing in general (Davenport *et al.*, 2020), and SMEs AI practices (Baabdullah *et al.*, 2021).

In business-to-business research, researchers proposed that AI may lead to improved market knowledge (Paschen *et al.*, 2019) and obtained some empirical support for a model of how AI leads to the creation of customer, user, and external market knowledge and that knowledge leads to rational decision-making which improves firm performance (Bag *et al.*, 2021). Researchers propose that the digitization of the business environment, inability to process increased amounts of information using traditional means, and interdependence of tasks within the company in compiling different information will lead to AI adoption by firms (Chen *et al.*, 2022). AI adoption in turn may lead to improved relationship quality (Chen *et al.*, 2022), innovation and performance (Sahoo *et al.*, 2024; Sullivan and Wamba, 2024; Chen *et al.*, 2022). Such positive performance outcomes were found to be contingent on the environmental dynamism (Sahoo *et al.*, 2024) and environmental hostility (Sullivan and Wamba, 2024). AI capabilities are also strongly related to several types of innovation and abilities to respond quickly to the environmental changes (Sahoo *et al.*, 2024; Sullivan and Wamba, 2024). Other contingencies may include how

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consulting companies, technology providers, and business partners interact to build various joint capabilities to implement the technology to serve all the stakeholder needs and to cocreate value (Johnston *et al.*, 2022; Li *et al.*, 2021). Additionally, researchers find that AI capabilities help B2B companies to better predict, mitigate and adapt to climate related changes in the business environment, make better business decisions and improve both business and environmental performance (Shankar and Gupta, 2024; Akter *et al.*, 2024). Despite these efforts to understand the importance of AI in business-to-business marketing, few studies have addressed the potential influence of AI technologies on the buying decisions inside the buying firm. A notable exception is an article by Johnston *et al.* (2022), where the authors note that AI innovations may shape the transformation of organizational buying centers (OBCs) and purchasing in general in the coming years. We build on the work of Johnston *et al.* (2022) and attempt to investigate more deeply and in detail *how* AI may shape the reconfiguration of the OBCs in industrial buying.

In doing so, we address the call articulated by Cabanelas *et al.* (2023), who, in their exhaustive systematic review of buying center literature, advocate for more research into the ramifications of the emerging technologies on the buying center concept. They underscore the pivotal role of technology within an evolving digital landscape, emphasizing the exigency for a meticulous examination of its impacts on decision-making processes and organizational frameworks. Researchers note that much of the value co-creation involves modern technologies and there is an urgent need to address the challenges of incorporating modern technologies in the decision-making process (Cabanelas *et al.*, 2023). They specifically point out that there is a gap in the existing literature in exploring how artificial intelligence can transform the OBC concept and what roles may emerge because of deployment of this innovative technology in the emerging OBC (Cabanelas *et al.*, 2023). Advances in information technology come in waves and tend to build on each other in affecting change in business processes including those in the OBC. For example, researchers view AI technologies as another wave of digitalization of business processes and decision making that could potentially make a significant impact on OBC structure and dynamics, but extant research has not sufficiently addressed the implications of such impact (Ehret *et al.*, 2024). Importantly, we examine the impact of AI adoption on OBC structure, dynamics, and the OBC outcomes because all these elements are critical in understanding the net effects of technology adoption in organizations (Cabanelas *et al.*, 2023).

Another important driver of changes in the buying center is the increased uncertainty of organizational buying (Ehret *et al.*, 2024). Prior research examined the role of information gathering and information processing by OBC to reduce the uncertainty of various B2B buying situations (Evald *et al.*, 2023; Johnston and Lewin, 1996; Henthorne *et al.*, 1993). In addition to obtaining information from traditional sources such as members of the selling centers (Siemieniako *et al.*, 2023; Zolkiewski *et al.*, 2023) and other collaborators (Evald *et al.*, 2023), OBCs are increasingly turning to other, primarily digital sources of information such as online customer reviews, ratings, and recommendations (Steward *et al.*, 2018) online professional communities and conversations taking place in the digital domains (Gustafson *et al.*, 2021, Diba *et al.*, 2019). Additionally, information gathered by sensors, actuators and other Internet of Things (IoT) devices is becoming increasingly important in information gathering by OBC (Osmonbekov and Johnston, 2018). Given this increasing information gathering activities and dramatic rise in the volume of information that is hard to combine and process by limited information processing capabilities of humans, companies are increasingly turning to building AI systems to help them to organize and synthesize information (Chen *et al.*, 2022). However, extant literature does not address these new developments in the context of OBC structure and processes.

To address these gaps in extant research, **we** focus on the potential influence of AI on various buying situations, such as straight rebuy, modified rebuy, and new task. Additionally, we explore the possible impact on the structure and dynamics of the OBC. Regarding OBC structure, we investigate how AI adoption might affect the size of the OBC and the degree of vertical and lateral involvement within it. As for OBC dynamics, we anticipate that the primary impacts will be on the levels of conflict and coordination among its members. We further propose that the changes in OBC structure and dynamics will lead to improvements in the efficiency and effectiveness of organizational buying. Therefore, we explore the following research questions:

RQ1. What are the probable effects of AI adoption on organizational buying situations?

RQ2. What is the impact of AI adoption on OBC composition, structure, and dynamics?

RQ3. What are the efficiency and effectiveness outcomes of AI adoption on organizational buying?

The rest of the paper is organized in the following way. We commence by reviewing AI definitions and AI system components and their relevance to industrial buying. Then we review the possible impact of AI on different buying situations. Next, we propose a set of testable propositions outlining the impact of the AI on OBC structure and dynamics. Finally, we discuss implications of this research for researchers and practicing managers.

Artificial intelligence

AI has been defined as "a system's ability to correctly interpret external data, to learn from such data and to use those learnings to achieve specific goals and tasks through flexible adaptation" (Kaplan and Haenlein, 2019, p. 15). **While many other definitions exist in the literature, it is worth mentioning a** definition from a **practitioner perspective**: AI is "application of mathematics and software code to teach computers how to understand, synthesize, and generate knowledge in ways similar to how people do it." (Andreessen, 2023). AI can do such things by employing a combination of technologies such as machine learning, natural language processing, deep learning, neural networks, physical robots and robotic process automation (Davenport, 2018).

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AI technologies typically involve a combination of vast amounts of data, enormous amounts of computing power via the use of supercomputers or a large network of connected computers, and sophisticated computer code or algorithms that direct computers to discover patterns in the data or "learn" the data fed to the system by the human "trainers." For example, a widely known ChatGPT chatbot created by OpenAI uses text data from publicly available sources such as the Internet, books, and articles. ChatGPT uses computing power owned or leased by OpenAI and went through an intensive period of "training" for months during which period the algorithms are tweaked for better performance of recognizing the patterns in the data. The current version is Generative Pre-trained Transformer (GPT) 3.5 uses Machine Learning (ML), where the system sifts through the data by itself and learns all the patterns and nuances of the data and considered to be Large Language Model (LLM) because its training data was limited to text. The current GPT 3.5 model also uses Natural Language Processing (NLP) and has 175 billion parameters which allows this piece of software to carry on human like text conversations with its more than one hundred million users worldwide, at the time of this writing. ChatGPT is also classified as generative AI because it can generate added information based on all the information it already acquired and classified using its 175 billion parameters. Of course, AI systems are not limited to text as technologists and researchers are constantly working on and improving AI systems that deploy images, video, sound and speech as data inputs. Panera Bread uses a database of their customers' voices for their voice recognition AI system and uses a database of customers' palm prints as well for customizing their orders (Fortune, 2023). A similar palm print system is deployed in Amazon's Whole Foods stores for checkout and package returns and one could expect customer biometric authentication to become more widespread for many reasons, including building customer databases to feed into AI systems. **In the context of business-to-business relationships, similar technologies are now also deployed to facilitate the sales function to serve the needs of various stakeholders in the value creation chain and their implementation presents unique opportunities and challenges** (McClure *et al.*, 2024).

It is important to note that AI builds on previous waves of innovation such as fast and connected computers, cloud storage for vast data storage, and sophisticated algorithms developed to process the data. Since data is so important in training AI algorithms and its functioning, some researchers and practitioners say, "data is the new oil" (Varian, 2018). Companies that had developed proper processes in gathering and storing data regarding their customers, suppliers, product usage and other types of data essential to businesses are more likely to develop effective AI systems. Firms that have implemented technology solutions that capture data using various Internet of Things (IoT) technologies and devices may also be ahead of competition as they have ready access to more data and more granular data from multiple touchpoints in their products and services (Osmonbekov and Johnston, 2018). Importantly, sophisticated technology users may have more variety in data sources than less sophisticated technology users which may prove essential in deploying more effective AI systems. For example, a firm using data from only from public sources, such as Internet and social media, may not find AI insights as useful as a firm using public sources in conjunction with internal company data, such as customer relationship management (CRM) or partner relationship management (PRM) data, and data from IoT devices installed throughout the value chain of a company and partner companies. Additionally, a proper implementation of AI systems requires much cooperation among business partners involved which may include technology providers, data providers, customers, suppliers and other stakeholders (Li *et al.*, 2021). Such technology transitions often provide a foundation for reengineered business processes for maximum efficiency and effectiveness of organizational buying (Johnston *et al.*, 2022).

Next, we examine the literature on industrial buying to make meaningful connections between AI and industrial buying structure and processes.

Nature of industrial buying

Classical industrial research described the industrial buying process as several sequential steps including recognition of a need, technical specification, supplier evaluation, and final purchase decision (Robinson *et al.*, 1967). Although these steps are theoretically simple, procurement in the real-world situations is subject to substantial variability because of differences in procuring diverse types of industrial products, buying situations, and interactions among members of OBC (Vyas and Woodside, 1984). Prior research in industrial buying investigated the distinct types of decisions faced by the industrial buyers (Doyle *et al.*, 1979; Roy and Terry, 1977; McQuiston, 1989; Moriarty, 1983). Previous research emphasizes that buying situations differ in terms of the complexity of the decision to be made, the significance of the buy to the organization, and their newness to the company (McQuiston, 1989). Many factors change across different buying situations. For example, earlier research discovers that the size of the OBC and its composition, the number of suppliers contacted during the search process, negotiating tactics, length of time to reach a decision and postpurchase evaluation all differ across buying situations (Evald *et al.*, 2023; Doyle *et al.*, 1979). Over the last 60 years, researchers have had varying foci in studying industrial buying behavior, from investigating individual transactions, to buying situations, various influences including responses to marketing efforts, to investigating dyadic and network relationships and finally customer journeys (Purmonen *et al.*, 2023; Steward *et al.*, 2019).

Since OBCs are the defining feature of industrial buying (Cabanelas *et al.*, 2023; Johnston *et al.*, 2022), the structure and processes of the OBCs has received considerable attention in the past research (Cabanelas *et al.*, 2023; Barclay, 1991; Dawes *et al.*, 1998; Johnston and Bonoma, 1981; McQuiston and Dickson, 1991; Patchen, 1974; Ronchette *et al.*, 1989; Johnston and Lewin, 1996). OBCs are fundamentally an uncertainty reduction and purchasing risk managing mechanism for organizations (Evald *et al.*, 2023; Johnston and Lewin, 1996). The dynamics of OBC composition and size changes are closely related to purchase risk. For example, as the purchase risk increases, the OBCs may include more higher-level managers (Johnston and Lewin, 1996) and focus on brand

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information in order the reduce risk in buying (Zablah *et al.*, 2010), especially in situations of extremely high or extremely low purchase importance (Brown *et al.*, 2012). Additionally, OBCs collect more information from various sources to reduce the risk (Evald *et al.*, 2023; Henthorne *et al.*, 1993), and communication and cooperation between OBC and selling team increases as the purchase risk increases (Evald *et al.*, 2023; Johnston and Lewin, 1996). In the next section we examine the different buying situations faced by industrial organizations and varying in degree of purchase risk and advance propositions on how AI adoption may impact these situations.

AI and business buying situations

Classical research in industrial marketing delineated three distinct buying situations faced by businesses: 1. Straight rebuy 2. Modified rebuy, and 3. New task (Robinson *et al.*, 1967; Hill and Hillier, 1977; Nicosia and Wind, 1977). Organizational buying theorists observe that buying situations differ in complexity and the types of buying processes that arise during the industrial buying process. For instance, in terms of time duration, in the new task buying situation companies take much longer to go through the stages of the purchasing process than in the straight rebuy situation (Evald *et al.*, 2023; Wind and Webster, 1972). Additionally, the size of the OBC is larger and the lateral involvement is greater in the new task situations than in the straight rebuy situations (Evald *et al.*, 2023; Nicosia and Wind, 1977). Typically, in the straight rebuy situation, the size of OBC stays the same throughout the process in contrast to the new task situation where size fluctuates as people come in and out of OBC depending on the stage of the process (Evald *et al.*, 2023; Doyle *et al.*, 1979). Finally, the post-purchase evaluation stage for the process is much more extensive for the new task situation and very brief, often only verbal for the straight rebuy situation (Doyle *et al.*, 1979). The interactions between OBC and the selling team also varies among the buying situations. For example, in the straight rebuy situation the purchasing function is typically the only one interacting with the sales team, while in a new task situation, technical personnel such as engineers are more likely to interact and cooperate (Evald *et al.*, 2023; Wind and Webster, 1972). In other words, the relative influence of the participants of the OBC vary across the three buying situations (Lewin and Donthu, 2005). The three buying situations represent different points on the risk/uncertainty continuum (Johnston and Lewin, 1996). High uncertainty typical of the new buy situation leads to more participation in the OBC and the expanded relative influence of the purchasing function (Spekman, 1979; Spekman and Stern, 1979). Straight rebuys tend to be recurring purchases for which most firms have established specific purchase procedures, but for new task situation where the uncertainty is high these formalized procedures may not be as applicable (Johnston and Lewin, 1996).

The straight rebuy is the simplest purchasing situation often involving maintenance, repair, and overhaul (MRO) items when a purchasing manager places an order to replace depleting stocks (Doyle *et al.*, 1979). Digital technologies have made these transactions much more efficient than before and may become fully automated in the future (Cabanelas *et al.*, 2023; Osmonbekov and Johnston, 2018). Since these types of purchases occur regularly and frequently, most companies may have copious amounts of stored data about straight rebuys which could be used to train AI systems. Once AI learns the data, it may suggest different buying, delivery, and maintenance schedules than currently used based on the data fed to the system. These may be optimized for price, inventory levels, production schedules, or combination of these and other factors that may be idiosyncratically important in different industries and individual firms. Consequently, firms may achieve more efficient or more effective purchasing processes using AI. However, if additional external data is fed to the AI system the situation may change. For example, social media data from companies like Microsoft's LinkedIn professional networking application is thought to influence all stages of organizational buying process (Diba *et al.*, 2019) and an influential new source of information for OBCs (Gustafson *et al.*, 2021). Another potential external source of information that could be fed into an AI system is online customer reviews, ratings, and recommendations that were found to have a considerable influence on B2B buyers (Steward *et al.*, 2018). It is entirely possible that an AI system could suggest alternative suppliers for products and services or some modifications to the past orders based on its understanding of the patterns in the vast amount of data it analyzed from both internal and external sources of information that were hard to analyze before due to the sheer volume of the information (Chen *et al.*, 2022). Depending on the validity of these suggestions (judged by the purchasing manager) the buying situation of straight rebuy may turn into a modified rebuy.

In modified rebuy and new task situations, OBCs larger than one person may be necessary to form to contact new suppliers, solicit proposals, and evaluate sales presentations. These activities could also be aided by AI as publicly available data from the supplier web sites, online customer reviews, ratings, recommendations (Steward *et al.*, 2018), and professional social media (LinkedIn) conversations (Gustafson *et al.*, 2021; Diba *et al.*, 2019) about the suppliers could be fed into the AI system to provide OBCs with the additional independent source of information. Both modified rebuys and new task situations are characterized by much more uncertainty and purchase risk than straight rebuys, so typically OBCs seek more information from a variety of sources to reduce uncertainty (Evald *et al.*, 2023; Johnston and Lewin, 1996; Henthorne *et al.*, 1993). Another path to reduce uncertainty is to build trust (Gao *et al.*, 2005; Morgan and Hunt, 1994) and humans would probably still prefer to interact face to face with other humans to build that trust (Nicholson *et al.*, 2001), similar to what happens at industry trade shows (Henthorne *et al.*, 1993), although professional social media interactions are increasingly becoming a close substitute (Gustafson *et al.*, 2021). The question remains on how much would OBCs trust the input of AI into the purchasing decisions. AI researchers argue that humans will never fully trust AI the way they trust other humans because computer minds are incomprehensible to humans and AI is incapable of any feeling or any concern for people's feelings (Gray, 2017).

Role theory (Biddle, 1979) may predict the emergence of new positions or roles to mitigate such concerns and improve the perception of trust in the new technology's input into the buying process. OBCs are social groups embedded in the

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context of organizations and the members of OBC perform distinct functions to meet behavioral expectations of the group and organization as a whole (Cabanelas *et al.*, 2023; Bachkirov, 2019; Wind and Webster, 1972). As AI empowered organizational purchasing becomes more accessible, it may be necessary to include data scientists into the existing OBCs. Data scientists are professionals who are trained in AI technologies and understand how AI reaches the conclusions from the data fed into the system. The data scientists' role and correspondent expectations will be to explain the inner workings of the organizational AI system to the members of OBCs, serve as trust transfer agents, and provide a human face to the abstract, non-human technological inputs into the organizational buying process. Data scientists thus may become an additional influencer in the OBC along with other technical personnel such as engineers, chemists, etc. This new influencer role in the OBC may be consistent with the notion that "novel technologies favor the appearance of new roles" (Cabanelas *et al.*, 2023, p. 76) in the organizational buying process that received some empirical support in past research about OBC roles (Raghavan *et al.*, 2020). Researchers observed similar phenomena in the previous waves of technology adoptions. For example, in the adoption of several types of complex business software in the past the Chief Information Officer (CIO) emerged as a gatekeeper and a powerful influencer in OBCs (Raghavan *et al.*, 2020). However, as technologies evolved over time and became easier to understand, implement and use, the role of the CIO in the OBC significantly diminished and the role of users became more important as they become more knowledgeable and do not rely as much on the CIO's input (Raghavan *et al.*, 2020). In a similar vein, the data scientist's role may be critical in overcoming the trust deficit at the earlier stages of AI adoption. However, as OBCs learn more about AI, and become more comfortable with the technology itself, and with the types of inputs that AI provides into the purchasing process, the data scientist's role as a trust transfer agent, and an expert may diminish over time.

To summarize our previous discussion, we advance the following proposition:

P1. Adoption of AI technologies will lead to: a) In straight rebuy situations, companies purchasing processes will become more efficient and effective, but may turn into modified rebuys depending on the information fed into the system; b) In modified rebuy and new task situations, data scientist's role will be important at the beginning of the adoption as trust transfer agent and influencer and will diminish over time.

It is important to understand that the impact of AI will not be limited to buying situations. It is likely that the structure and processes of the buying center itself will be altered under the influence of this technological trend (Cabanelas *et al.*, 2023; Johnston *et al.*, 2022). The impact is discussed in the next two sections.

AI and buying center structure

Given the topic of AI technologies' impact on procurement, our research focuses on the ability of buying center members to gather, evaluate, and exchange information necessary for purchasing decisions. Organizational researchers argue that companies can adjust their organizational design to match the level of uncertainty they face to achieve optimal performance (Galbraith, 1974). Two strategies are suggested: preplanning all activities to reduce uncertainty or being ready to adapt swiftly to changing environmental conditions (Galbraith, 1974). The first strategy requires decision makers to acquire sufficient information in advance of task execution (Wang *et al.*, 2013), while the second strategy demands flexibility and the ability to process information rapidly and act upon new insights. Previous research suggests that Internet applications enable companies to pursue both strategies more efficiently by facilitating the gathering of ample information in a timely manner compared to older technologies (Saeed *et al.*, 2005). IoT technologies are also expanding the scope and scale of information gathering beyond human reporting of data (Osmonbekov and Johnston, 2018; Porter and Heppelmann, 2014). If deployed effectively, AI has the potential to further enhance organizational information processing capabilities and empower decision-makers to navigate uncertainty and recent research provides some empirical evidence to support this view (Paschen *et al.*, 2019; Bag *et al.*, 2021; Chen *et al.*, 2022). Such effects of modern technologies will be operational both at the organizational level as well as at the level of OBCs.

The concept of an OBC refers to a collective of organizational actors within a firm who initiate, evaluate, and finalize purchasing decisions (Cabanelas *et al.*, 2023; Wind and Webster, 1972). Previous studies in industrial marketing have explored how the structure of the OBC varies under different conditions (Moriarty, 1983; Johnston and Bonoma, 1981). Researchers have observed that when a decision is surrounded by higher levels of environmental uncertainty, the OBC requires more information and encourages information sharing within the group (Spekman and Stern, 1979). Additionally, the behavior of the OBC is influenced by factors such as the decision stage, product type, and unique characteristics of the companies involved (Lilien and Wong, 1984; Johnston and Bonoma, 1981). As we discussed it in the previous sections, the type of buying situation plays a crucial role in shaping the structure of the OBC (Johnston and Bonoma, 1981) and AI may lead to the reorganization of OBCs structure and processes (Cabanelas *et al.*, 2023; Johnston *et al.*, 2022). According to the referenced buying center literature, the primary structural characteristics identified are the size of the OBC, and lateral and vertical involvement in the OBC.

Organizational buying center size

Previous studies indicate that the primary factors influencing the size of the OBC are the complexity of the purchase process (Cabanelas *et al.*, 2023; Johnston and Bonoma, 1981) and the level of formalization (Dawes *et al.*, 1998). In cases where the decision-making process is intricate and involves extensive information gathering, evaluation, and analysis, the OBC tends to expand in size to accommodate the additional personnel required for communication and transactional tasks (Johnston and Bonoma, 1981). Formalization refers to the degree to which the activities of the OBC are governed by rigidly prescribed rules and procedures (Lau *et al.*, 2003; Johnston and Bonoma, 1981) and is related to

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the degree of formalization in the corresponding company overall (Wood, 2005). Greater formalization leads to increased amount of paperwork, which, in turn, necessitates a larger number of personnel involved in the buying decision (Dawes *et al.*, 1998). The adoption of AI technologies is likely to impact both the complexity and formalization drivers, influencing the size of the OBC.

The complexity of many industrial buying decisions may be reduced by the thoughtful and appropriate deployment of AI systems in industrial buying. Companies have significantly reduced labor-intensive data gathering procedures by using web-based tools (Osmonbekov *et al.*, 2002), and IoT technologies provide decision makers with vast amounts of relevant data for industrial purchasing (Osmonbekov and Johnston, 2018). The maturation and advancement of AI may further reduce the complexity of industrial buying decisions as it provides analysis and insights from the large amount of digitized data generated by the previous waves of technological development in industrial buying (Ehret *et al.*, 2024). In fact, increased digitization is one of the drivers of the adoption of AI by companies (Chen *et al.*, 2022). Importantly, AI can synthesize vast amounts of data from many independent sources such as internal PRM databases (Osmonbekov and Gruen, 2013), supplier website data, and include data from professional social media such as LinkedIn (Gustafson *et al.*, 2021; Diba *et al.*, 2019), and online customer reviews, ratings and recommendations (Steward *et al.*, 2018). Such synthesis is beyond the abilities of any individual or even a large group of individuals and provides a major advance in the information processing capabilities for the OBC. Recent studies suggest that AI adoption may lead to increased customer and market knowledge because of such synthesis of information capabilities of AI technologies (Paschen *et al.*, 2019; Bag *et al.*, 2021).

Digitization of the industrial buying and deployment of digital tools and data have led to reduced formalization of the buying processes (Cabanelas *et al.*, 2023; Lau *et al.*, 2003; Osmonbekov and Johnston, 2018; Osmonbekov *et al.*, 2002). AI technologies are highly computerized and may likely to further reduce the formalization of the buying process. While decision makers already have access to a lot of purchasing related information, sophisticated AI technologies are able provide decision makers with recommendations based on objective ML based analysis of thousands, millions or potentially even billions of data points, depending on the availability of data that feeds into the AI system. As previously discussed, such systems may consider billions of parameters and such systems provide solutions to problems that may be completely outside of human consideration and flexible depending on the decision at hand and the data that is used by the system. The nuanced analysis of so much data may no longer be compatible with rigid rules that have been established previously based on the traditional data gathering and analysis. Consequently, formalization of OBC decision making may be reduced to accommodate the AI driven industrial purchasing processes.

Viewing it from the perspective of information processing, the utilization of technology for collecting and analyzing data related to purchasing criteria significantly enhances the information processing capacity of the buying center. Without such software, humans would be overwhelmed by the sheer volume of raw data (Chen *et al.*, 2022). These software tools effectively present decision-makers with information that is easily digestible and can be meaningfully utilized. Generative AI can write purchasing proposals, justification for selecting certain suppliers, and draw out purchasing schedules based on the analysis of the data much more efficiently than human analysts in the OBC. Consequently, the improved information processing capabilities within the buying center may result in a reduction in its size, as a smaller number of individuals can analyze and process a larger amount of information. On the other hand, as mentioned in the previous section of buying situations, data scientists may be more involved in the OBCs, at least at the first stages of adoption of AI. Since there is less need for data collection clerks and analysts but an increased need for data scientists to deploy AI and instill trust in the AI recommendations the net effect may be reduced number of participants in the OBC.

To summarize, as the complexity of the purchase process decreases, formalization levels decline, and AI-enabled procurement processes improve information processing capabilities, it is likely that the number of individuals engaged in the buying decision will decrease.

Organizational buying center involvement

OBC theorists distinguish between two types of involvement: vertical and lateral (Cabanelas *et al.*, 2023). Vertical involvement is the number of hierarchical levels involved in the purchasing decision (Johnston and Bonoma, 1981). The formal position held by an employee significantly influences how they obtain and utilize procurement information (Ronchetto *et al.*, 1989). The hierarchical positions of OBC members can range from top-level senior managers to plant managers, production and maintenance technicians, and clerks. Researchers have observed that the involvement of buying center members in various procurement stages varies based on their formal rank (Choffray and Lilien, 1978). Lateral involvement is the number of different departments or functional areas involved in the purchasing decision (Cabanelas *et al.*, 2023; Evald *et al.*, 2023; Johnston and Bonoma, 1981). Within the OBC, individuals usually come from different departments and functional areas and may receive different compensation, which in turn affects their respective roles in the OBC (Anderson and Chambers, 1985).

Companies' inability to process the vast amounts of available digitized information using traditional means are increasingly driving them to adopt AI technologies (Chen *et al.*, 2022) and it enables them to increase their customer, user and market knowledge and use more rational decision making (Bag *et al.*, 2021). Enhanced rational decision-making and greater knowledge are expected to empower firms to optimize their purchasing processes, aiming to reduce the involvement of senior managers – individuals whose time most valuable – in procurement decisions. Prior research on the impact of information technology on OBC vertical involvement suggests a similar dynamic of flatter hierarchical structure of OBC (Osmonbekov and Johnston, 2018) and less politicized information processing with the OBC (Cabanelas *et al.*, 2023). With AI in particular, generative AI is able produce reports using all the data fed into the system targeting every stage of the

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industrial buying process. Importantly, these reports could be easily customized using simple language commands to AI system to use the language and goals relevant to different hierarchal levels within the company. For example, report for senior executives will include justifications based on the strategic priorities of the firm, assuming top management articulated those priorities in written form, and more efficiency type justifications for unit and production floor management. Such customized AI generated purchase justifications may make communication within OBCs more efficient and reduce vertical involvement in the OBC.

As for the lateral involvement, the involvement of data scientists will increase the size of the OBC, as discussed in the previous section and incrementally increase the lateral involvement of the OBC, as the data science department or functional area will be involved in the OBC. However, the reduction of the other type of personnel, such as the clerks and other data collectors may result in the net reduction of the functional areas involved in the OBC. It is important to note that the level of collaboration among customers, suppliers, technology providers, and consultants in the implementation of AI technologies will have a large bearing for the outcomes for the company in general (Li *et al.*, 2021) and lateral involvement in OBC. For instance, if during the implementation of AI all functional areas actively participate and contribute all available data for the training of AI and its proper integration with the existing internal and external data systems, we could expect the lateral involvement in the future OBCs **to** be reduced. On the other hand, if various departments withhold relevant data during the implementation of AI, deliberately or otherwise, or data sources are not well integrated, we could expect the lateral involvement in OBC remain the same. Such contingency perspective is in line with prior research on AI in organizations where positive organizational outcomes are contingent on the broad business environment (Sahoo *et al.*, 2024; Sullivan and Wamba, 2024), and how technology consultants, technology providers and other stakeholders implement technology collaboratively to create value (Johnston *et al.*, 2022; Li *et al.*, 2021). We propose that such collaborative atmosphere is also necessary within the company, among various departments that contain and own information pertinent to the purchasing decisions and could be integrated into the company's AI systems.

Based on the discussion above we advance Proposition 2:

P2. Adoption of AI will impact OBC structure in the following way: a) OBC size will be reduced because of the reduced complexity of buying, reduced formalization and increase in information processing capabilities of the buying center; b) Reduce both vertical and lateral involvement, contingent on the proper integration of data sources and implementation of AI system.

Organizational buying center dynamics

According to previous studies, the behavioral and attitudinal aspects of the OBC undergo transformations in response to changes in the business environment (Johnston *et al.*, 2022). The adoption of AI could serve as a catalyst not only for changes in the OBC's structure but also for the underlying dynamics (Cabanelas *et al.*, 2023). Our proposition is that these changes will impact the conflict and coordination dynamics of the OBC. As mentioned previously, the buying center consists of members from various departments within the firm. Consequently, due to the diverse viewpoints and perspectives brought to discussions, interdepartmental conflicts often become inevitable (Cabanelas *et al.*, 2023; Barclay, 1991; Ding, 1997). Moreover, differences in problem-solving approaches and management styles frequently lead to misunderstandings, anxiety, and mistrust (Ding, 1997). Although many factors contribute to conflict, Barclay (1991) suggests that the primary reason for interdepartmental conflict is the presence of communication barriers between departments, as people from different departments and hierarchical levels may bring their own logic, priorities and mindsets to a given buying situation (Evald *et al.*, 2023; Wood, 2005). Prior research also identified potential mitigation strategies for OBC conflict such as understanding the perceptual differences among the participants and developing targeted programs (Kiser *et al.*, 1975), development of leadership (Garrido-Samaniego and Gutiérrez-Cillán, 2004; Kohli, 1989) and negotiating skills (Tellefsen, 2006). It has also been suggested that new organizational technologies have the potential to reduce OBC conflict because the information processing may become more objective and less political (Cabanelas *et al.*, 2023; Osmonbekov and Johnston, 2018).

The adoption of AI technologies has the potential to mitigate conflict in the buying center by enhancing communication among its members. By providing insights and recommendations gleaned from troves of objective data from a variety of disinterested data sources, AI may help reduce the divergence in viewpoints during purchasing discussions. Furthermore, these discussions might become less political and more objective, as the data's credibility is less likely to be questioned when obtained from disinterested and unbiased sources. Recent empirical findings on AI in industrial buying suggest that with AI implementation the decision making has become more informed and more rational (Bag *et al.*, 2021). However, it is important to acknowledge that social and political factors in decision-making may not completely disappear. Instead, the availability of objective data could increase its utilization and make the process less political, leading to an incremental and gradual reduction in conflict.

Coordination involves effectively managing the interdependencies among various resources and activities within a system (Malone and Crowston, 1994). In the context of the OBC, the size of the group dictates the need for proper management of the complex interactions and interdependencies among its members (Pedersen, 2023; Johnston and Bonoma, 1981). Even though different members of the buying center may have specific tasks and responsibilities, they must inevitably coordinate their diverse activities and tasks, to reach a collective decision (Pedersen, 2023). For example, while a purchasing manager might be responsible for comparing prices from different suppliers, an engineer may be tasked with assessing and comparing technical specifications (Osmonbekov and Johnston, 2018). Throughout this information processing, data needs to be shared and communicated among the various members of the OBC. However, communication often occurs through interpersonal

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dyads rather than in a group setting (Johnston and Bonoma, 1981) which could be inefficient from the coordination perspective.

Viewed through the lens of information processing, advanced information technology enhances an organization's capabilities in handling information (Chen *et al.*, 2022) and facilitates improved coordination both within and between organizations (Pedersen, 2023; Malone and Crowston, 1994). Similarly, the adoption of AI technologies can aid in managing the intricate web of communication and information sharing within a buying center. For instance, generative AI can provide reports to all OBC members at the same time and respond to queries that use normal human language. This grants everyone equal access to the knowledge amassed in the system, allowing them to compare purchase requirements with the actual performance of the product in the past and conversations about the product held in the present on the professional social media platforms by competitors and other customers and suppliers. Additionally, AI is able perform many tasks previously performed manually by purchasing managers and engineers separately and independently, as mentioned in previous paragraph. Since AI performs them simultaneously, the need for coordinating, in this instance, the tasks of an engineer and a purchasing manager becomes no longer necessary. The enhanced coordination also stems from OBC members being able to share knowledge with greater speed and ease, due to the AI technology. It is worth mentioning again that enhanced OBC coordination is contingent on the ability and willingness of a firm to integrate all available organizational data pertinent to purchasing decisions into the AI system of the firm.

We advance the following proposition to summarize our discussion:

P3. Adoption of AI will impact OBC dynamics in the following way: a) Reduce conflict because more objective information is available, making decisions more rational and data driven; b) Increase coordination because AI technologies enhance information processing and reduce interdependent tasks.

Industrial buying outcomes

Industrial purchasing outcomes are commonly assessed in terms of efficiency and effectiveness, as indicated by previous research (Cabanelas *et al.*, 2023; Osmonbekov *et al.*, 2002). Efficiency refers to reduced expenditures and quicker procurement cycle times, while effectiveness pertains to higher-quality buying decisions within organizations (Cabanelas *et al.*, 2023; Osmonbekov *et al.*, 2002). The purchasing function in organizations is constantly striving to minimize the use of resources and time in making decisions as their ability to do so may affect the costs of design and manufacturing (Cabanelas *et al.*, 2023; Johnston and Bonoma, 1981). It requires balancing of and solving for multiple and complex tradeoffs among a range of factors that are considered in purchasing such as price, transactions costs, delivery, customization, and service levels, among others (Pedeliento *et al.*, 2018; Dadzie *et al.*, 1999), as well as reaching compromises among different members of OBC (Tellefsen, 2006). The integration of AI technology into industrial purchasing processes is expected to enhance both efficiency and effectiveness in procurement. This improvement can be attributed to two main drivers. Firstly, the technological impact of AI, as advanced technology typically leads to enhanced efficiency and effectiveness in companies (Chen *et al.*, 2022; Bag *et al.*, 2021). Secondly, organizational changes in response to the adoption of AI technology are likely to influence the efficiency and effectiveness of decision-making in industrial purchasing.

Regarding efficiency, the optimized configuration of the buying center (achieved through a smaller size and flatter hierarchical level) can lead to cost savings. This is because companies can allocate their workforce more efficiently to other tasks, making better use of the limited resources of employee and manager time. Additionally, the enhanced coordination of OBC and incremental reduction in conflict will enable OBC to arrive at decisions more rapidly, further contributing to efficiency gains. Concerning effectiveness, access to AI generated insights regarding the product or service under consideration can empower the buying center to make higher-quality decisions. Making decisions based on partial or incomplete information often results in impulsive choices that could prove to be costly overall. Having comprehensive information derived from conventional sources as well as AI generated insights positions the OBC to achieve superior decision-making quality. Furthermore, changes in the buying center can also contribute to increased effectiveness. For instance, a reduced level of conflict among members may lead to improved purchasing decisions, as the OBC collaboratively works together to address purchasing challenges. Additionally, the OBC may have more time to co-create value within the OBC as well with external parties such as suppliers and technology providers (Johnston *et al.*, 2022; Li *et al.*, 2021). Prior research suggests that AI adoption enhances the customer and market knowledge in firms and leads to more rational decision making (Bag *et al.*, 2021) and may lead to positive organizational outcomes such as improved relationship quality and financial performance (Chen *et al.*, 2022). We propose that similar processes may occur at the OBC level and improve both efficiency and effectiveness of purchasing decisions. Therefore, the following proposition is advanced:

P4: Adoption of AI by industrial buyers will enhance procurement effectiveness and efficiency by reducing information processing costs and facilitating higher-quality decisions through increased information availability.

Discussion

In this section we outline theoretical implications, implications for practicing managers, future research directions and limitations of our study.

Theoretical implications

This research answers the call by Cabanelas *et al.* (2023) for more research on the impact of artificial intelligence on the structure and dynamics of OBC. We address this gap in extant research by examining previous research in artificial intelligence and buying center and articulating several theoretical propositions that could be tested by future research. As AI could be considered another

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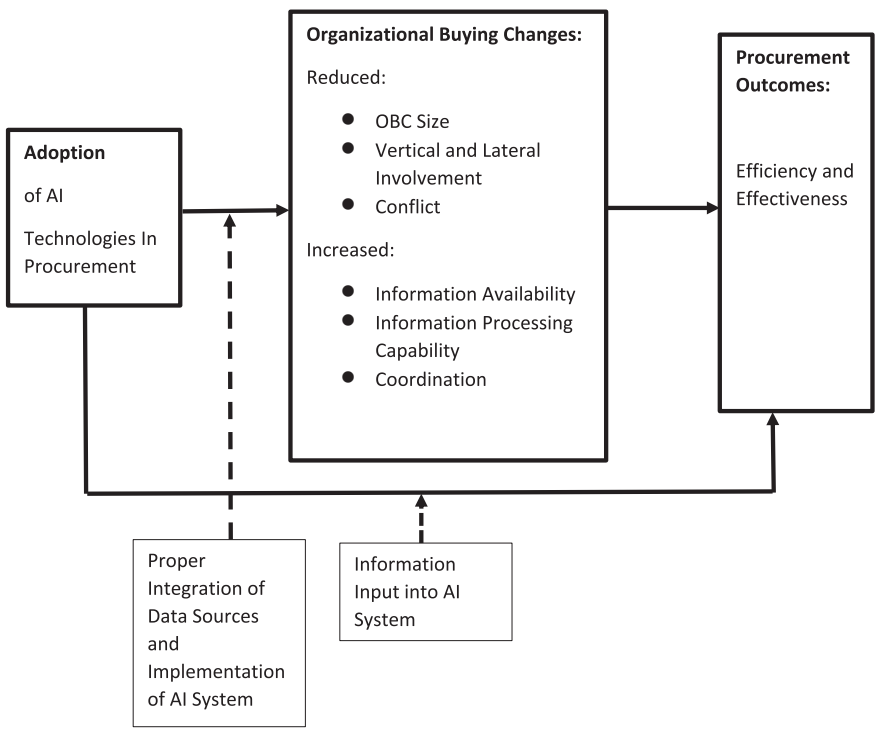
technological development building on previous waves of digitalization of business processes (Ehret *et al.*, 2024), we suggest that AI will enhance OBC capabilities in gathering and processing purchasing related information from diverse sources such as selling centers (Siemieniako *et al.*, 2023; Zolkiewski *et al.*, 2023), public partners (Evald *et al.*, 2023), online professional communities and conversations (Gustafson *et al.*, 2021, Diba *et al.*, 2019), and online reviews, ratings and recommendations (Steward *et al.*, 2018). As such sources of information, both internal and external, continue to proliferate by including machine gathered information via IoT devices (Osmonbekov and Johnston, 2018) the sheer volume of data may exceed human abilities to process these vast amounts of information in an efficient and effective manner (Chen *et al.*, 2022). In such situations, the deployment of AI systems in OBCs may become particularly important in facilitating an efficient and effective organizational purchasing decisions.

Our theoretical model (Figure 1) suggests that the adoption of AI in organizational buying will have both direct and indirect effects on its efficiency and effectiveness.

The direct effect is theoretically straightforward and broadly supported by related extant research on AI. The information processing view suggests that any firm initiative that improves its capability to process information may result in better firm performance (Galbraith, 1974). For instance, researchers find that the deployment of AI systems improves firms' information gathering, and information processing capabilities (Chen *et al.*, 2022), market knowledge (Paschen *et al.*, 2019; Bag *et al.*, 2021), rational decision making (Bag *et al.*, 2021), ability to adapt to changing situations (Sullivan and Wamba, 2024; Shankar and Gupta, 2024; Akter *et al.*, 2024) and innovation (Sahoo *et al.*, 2024). We hypothesize that that these effects will be active at the OBC level and enhance OBC's efficiency and effectiveness in a direct way. The indirect effect is theoretically more nuanced and interesting than the direct effect. We hypothesize that the changes in OBC structure and dynamics will in turn affect the efficiency and effectiveness of OBC. In terms of structure of OBC, we propose that as AI increases information processing capabilities of OBC it may potentially decrease formalization, vertical and lateral involvement and the OBC size. In terms of OBC dynamics, we propose that the reduction of conflict and coordination within OBC because of more rational and data driven decision making and reduction in interdependent tasks. Improved structure and dynamics of OBC will in turn enhance the efficiency and effectiveness of organizational purchasing. In our view, the exploration of both direct and indirect effects of AI adoption represents an important contribution to the extant literature on the effects of AI on organizational buying.

Another theoretical contribution we attempted to make is to examine the impact of AI adoption on the composition of OBC. Previous research suggests that innovative technologies may influence OBC in terms of new roles in the OBC (Cabanelas *et al.*, 2023) and the relative influence of different members in OBC on the purchasing decisions (Raghavan *et al.*, 2020). Since AI is a novel and complex technology and the institutional imperative may dictate the incorporation of AI insights into the purchasing process (Chen *et al.*, 2022), role theory suggests the emergence of a new influencer role whose function may be to explain AI to other members of OBC and provide a trust bridge between this novel technology and

Figure 1 Theoretical model of the impact of AI advancements on procurement



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traditional OBC decision making. We hypothesize that data scientists may fill this role and fulfill the expectations of OBC in terms of incrementally developing trust toward potential insights derived from AI system. Additionally, we suggest that this new role may be diminished over time, as OBC become better acquainted with AI technology, trust it more, and no longer need expert guidance and counsel of data scientists. Such outcome would be consistent with previous research in technological adoption and functioning of OBC (Raghavan *et al.*, 2020).

Research directions

We would like to highlight several areas that may be particularly attractive for future research. In proposition one we suggest the possible impact of AI on different situations in purchasing. It would be interesting to investigate this proposition empirically. Proposition 1a suggests that some straight rebuy situations may turn into modified rebuys based on the insights generated by AI. It would be fascinating to examine what percentage of the decisions would be transformed into modified rebuy and how it would stretch the imaginations and existing decision frameworks of OBCs or purchasing managers. It would also be useful to examine the role of data scientists in AI enhanced purchasing. How much correlation would there be between the trust in the data scientist(s) and the AI purchasing input? How long would it take for the OBCs to get comfortable with the AI input into purchasing before the data scientist(s) involvement in the OBC becomes less necessary? Proposition two regarding the OBC size and involvement could also be tested empirically to understand the practical impact and size of the effect. Although we think directionally, we are correct, it would be worth knowing what the practical effect size is in the real world. Since Proposition 3 is about the dynamics of the OBC, we would expect future empirical research to produce conflicting results, as these processes are ever changing and need to be examined in a longitudinal context. For example, one could imagine the initial stages of AI adoption in purchasing to be confusing to OBC, with new informational inputs and added information processing capabilities. It is entirely possible that the initial stage of adoption would be more conflict prone, as members of OBC argue over how correct AI suggestions are and how much weight to put on the system's input. It is also possible that it could be less coordinated in the beginning, as some new processes are put in place that replace old processes and people. However, we would expect, eventually, the dynamics will work in the predicted direction, subject to empirical verification. Of course, what the long run is exactly is also an open empirical question.

Proposition four makes a straightforward prediction on efficiency and effectiveness impact of AI technology on organizational buying. Future research could explore moderators of this direct effect. For instance, since availability of large amount of data is required for the effectiveness of AI systems, it could be argued that AI impact may be limited to large organizations that tend to possess and own substantial amounts of historical purchasing and other data to become a part of a sophisticated AI system. On the other hand, small organizations may not have such data which could make investments in AI system less practical. Another consideration is of course the cost of the development of AI algorithms and data scientists involved in this development and training of the system. Although research suggests that adoption of AI by SMEs is beneficial to their performance (Baabdullah *et al.*, 2021), one would expect the size of the organization could be a moderator in the AI adoption to performance link. Another moderator of this important relationship could be the way AI is implemented. Implementation of technology has always been a sticking point in technology research and its impact on business performance. We do not expect AI technology to be an exception. Researchers address the question of AI implementation conceptually (Johnston *et al.*, 2022), and with qualitative research (Li *et al.*, 2021), thus providing future empirical investigations on the moderating role of technology implementation a much-needed foundation.

Limitations

There are several limitations of this research that could be addressed in future research as well. Social network analysis could be a useful theoretical tool to analyze OBC, since OBC represents a complex network of social actors tied by a web of communications and relationships (Pedersen, 2023; Johnston and Bonoma, 1981). It would be interesting to examine the changes in the communication patterns and power shifts within OBC as AI impacts the OBC structure and dynamics. For example, do data scientists become the most influential member of OBC with the highest centrality in the OBC social network? We do not employ social network analysis in this research, and it could be a promising future research topic for scholars interested in the AI's impact on organizational buying process. Another limitation of this research is that we narrow our focus to AI's impact on intra-OBC changes within a single company overlooking a concept of buying ecosystems that may include interorganizational networks and broader environmental factors (Ehret *et al.*, 2024). Additionally, in discussing the changes within OBC and buying situations, we are limiting our discussion to the traditional perspective of product purchases and do not address value co-creation in complex services (Zolkiewski *et al.*, 2023). The examination of AI's impact in the multi-phase, complex service delivery processes so common in today's B2B landscape provides another fertile avenue for future research. The domain of marketing analytics may experience a disruption because of proliferation of AI tools. AI may enhance the ability to process vast amounts of information and at the same time replace some labor-intensive analysis of data with machine learning and analysis. Such disruption also could be researched further by researchers and marketing analytics experts [1].

Managerial implications

While the long-term benefits of AI technologies for buying organizations may not be fully evident yet, several preliminary projections can be made. These technologies offer opportunities to attain cost efficiencies and make higher quality purchasing decisions. AI increases the information processing capability of OBC and therefore increases the efficiency of the buying process. We suggest that straight rebuy situations may become more efficient or may turn into a modified rebuy because AI may suggest alternative products and/or suppliers that humans have not thought of before. Herein lies an opportunity for "out"

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suppliers to come in and compete for the business and a threat for the "in" suppliers who may not be anticipating such competition. The 'in' suppliers should be prepared for this threat and increase in competition for a stream of rebuys they may have been accustomed to over an extended period of time. For example, providers of MRO products should be prepared to answer more detailed customer questions during the negotiations for the renewal of their contracts. As AI systems proliferate, customers will be increasingly armed by AI system's data driven insights to evaluate their existing suppliers more critically. In modified rebuy and new task situations, the buying center's information processing capabilities are also expected to improve through the adoption of AI tools. Since, humans may not initially trust AI input, the incorporation of credible humans, such as data scientists who design, train, and implement AI systems may be a crucial element in making AI work well for the companies and fulfill the promise of more efficient and effective purchasing using this advanced technology.

The integration of AI inputs into the purchasing processes of organizations holds several implications for firms marketing to them. First, with the anticipated reduction in buying center size, seller firms can implement more targeted selling strategies, focusing on a smaller subset of members. For example, similar to B2C firms targeting specific customers on social media using AI, B2B selling centers could target the smaller buying centers using AI on professional social media platforms such as LinkedIn. Secondly, seller firms should be particularly attentive to the influence dynamics within the smaller buying center. It is essential to cater to the needs and demands of each member as they now have more influence on the ultimate buying decision. Additionally, the flatter hierarchical structure of the buying center may imply a lower likelihood of having a single decider, such as a senior manager. More broadly, selling teams need to carefully consider the input that AI technology has on the customers' procurement processes. Specifically, what are the data inputs that the customers use for their procurement AI systems? Conversations on professional social media may become a standard data feed into such systems, so selling firms need to monitor these very carefully as these may be a crucial information input into purchasing decisions. For example, seller organizations may want to establish certain policies for all their employees, not just the selling centers, on how to engage in the professional social media platforms to maximize their chances of establishing new business relationships and maintain existing relationships. Because this data source may have a considerable influence on the purchasing decision of the buyer firm, it may be necessary for the selling team to establish social media war rooms, similar to what well established consumer brands have done to monitor consumer social media and intervene to protect their brand reputation. However, the purpose of these war rooms would be not the intervention, but careful monitoring of all product category information produced in the cyberspace, and constant creating of content that is favorable to the focal firm's products and services. To be effective in this AI-empowered purchasing era, selling centers would also want to have AI experts as part of the selling center. The role of data scientist in the OBC should be mirrored in the selling center by providing expertise and guidance to the selling center, just like the data scientist would do in the OBC.

Conclusion

Artificial intelligence technologies are likely to have a transformative impact on organizational buying behavior in the coming years and decades. In this research, we examine the potential impact of this revolutionary technology and conclude that it has the potential to make OBC a much more effective and efficient decision-making unit. The main driver of such potential is the enhanced information-processing capability of OBC as a unit of organizational decision-making. As OBCs become more capable of mastering the vast amounts of information pertaining to the buying choices, they will be able to reduce uncertainty and make increasingly better decisions, define and manage risks more appropriately, and use valuable organizational resources more efficiently and effectively. Such beneficial outcomes are contingent upon proper implementation of AI systems and full integration of all available data sources essential to organizational buying since data is the life blood for any AI system. Given the iterative nature of AI technology, it is evolving rapidly in terms of accuracy, speed, and the variety of data it can process. Organizations that could take advantage of this technological juggernaut would be positioned to achieve increased performance and productivity at OBC as well as the firm level.

Note

1. We thank an anonymous reviewer for this suggestion.

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